

Project Title

Rice Leaf Disease Classification and Mango Leaf Disease Classification Using Deep Learning

Students Information

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Project Description

This project focuses on developing and evaluating deep learning classification models for plant disease detection using leaf images.

Two separate image classification models were implemented:

- A model for **Rice Leaf Disease Classification**
- A model for **Mango Leaf Disease Classification**

The models are trained to classify healthy and diseased leaves based on visual symptoms appearing on the leaf surface.

The project includes:

- Data preprocessing and augmentation
- Deep learning model training and evaluation
- Performance analysis using evaluation metrics
- Experimental comparisons between different models
- Final predictions and visualization results

The goal of the project is to explore the application of deep learning techniques in agriculture to support early disease detection and improve crop management.

Datasets

- Rice Leafs Disease Dataset
- Mango Leaf Disease Dataset

GitHub Repository

<https://github.com/kclnk/SelectedTopics-DL/tree/main>

The Repository Contains

- Complete project source code
- Dataset information and preprocessing steps
- Model training and evaluation notebooks
- Trained models and generated results
- Performance metrics and visualizations
- Final documentation and reports